An Algorithm-Selection Matrix for Auditable Redistricting Workflows

Gio Della-Libera

Draft — May 12, 2026

Abstract

This draft defines a practitioner-facing matrix for choosing among construction, search, exact optimization, ensemble, Pareto, and audit paths. The central claim is methodological: algorithm choice should follow the claim type, data availability, required certificate, and RPLAN/RCTX audit surface. The matrix is a routing aid for auditable workflows, not a ranking of algorithms or a legal sufficiency test.

1 Introduction

U.12 is a methodology paper. It helps users choose between construction, search, exact, ensemble, Pareto, and audit workflows without implying that every workflow supports the same evidence or legal claim.

The paper takes U.0's evidence ladder as its premise: a valid plan, a benchmark result, an optimality certificate, an ensemble distribution, and a legal posture serve different purposes. Selection begins by naming the intended claim and then choosing the weakest workflow that can generate the necessary artifact trail for that claim.

2 Selection Matrix

The matrix maps task type to algorithm family, CLI surface, evidence requirement, and audit artifact. Construction methods provide baselines, search methods explore alternatives, exact methods can provide bounded certificates, multi-objective methods expose trade-offs, and audit methods verify declared artifact contracts.

Compositor Framing

The selection matrix is also a three-layer compositor checklist. For any recommended workflow, the operator should name the structure layer, weight layer, and search layer before interpreting the result. Changing one layer creates a new run recipe; it should not be described as an isolated algorithm comparison unless the other two layers are held fixed. The candidate object is therefore the resolved run recipe itself: structure choice, weighting choice, search mode, backend settings, and config hash. The decision rule is the row chosen for the intended claim, and the consequence is the artifact bundle that must be retained for audit.

Need	Preferred family	Required evidence	Audit handoff
Fast baseline	Construction	L0/L1 fixture and smoke tests	RPLAN with constructor metadata
Improve candidate	Local/LNS search	Validity-preserving move trace	RPLAN/RCTX run summary
Bounded claim	Exact optimization	Bound, witness, or solve report	Certificate and manifest
Distributional context	Ensemble/SMC	Protocol and sampled-plan record	Exported plans and context
Trade-off choice	Pareto selection	Frontier and selection rule	Selected-plan package
Artifact verification	Audit verifier	Declared profile pass/fail	Certificate bundle

Table 1: Method-family selection by claim need and audit handoff.

The matrix is conservative by design. If the claim requires a bound, a search heuristic can provide useful candidates but should not be described as exact. If the claim requires distributional context, a single optimized plan is an example, not a sample. If the claim is legal or policy-facing, the workflow must carry enough evidence to support interpretation without letting the audit certificate stand in for that interpretation.

Preflight Questions

Before selecting a workflow, the operator should answer four questions: What claim is the output meant to support? What evidence level is required for that claim? Which CLI surface or crate currently produces the relevant plan or certificate artifact? Which RPLAN/RCTX sidecars must be retained so another reader can inspect the run?

Evidence Fields

A publication-ready selection record should include the resolved structure, weights, search mode, seed rule or seed budget, objective, selected artifact, RPLAN/RCTX identifiers, sidecar list, certificate or solve-report path when available, and the claim boundary stated in plain language. This prevents a valid audit certificate from being read as an optimality proof, a local-search improvement from being read as a global bound, or a percentile seed run from being read as an ensemble over all valid plans.

3 Audit Boundary

The RPLAN/RCTX fixed point is the common handoff. Selection guidance should say which sidecars, manifests, certificates, or run summaries are required before a workflow can support a stated claim.

The audit boundary also prevents category mistakes. A certificate can show that a declared verifier accepted a plan package, or that an exact method reported a bound under stated assumptions. It does not make an ensemble representative, make a heuristic globally optimal, or make a legal posture sufficient outside the recorded profile.

4 Limitations

The matrix is a decision aid, not a proof that any chosen workflow is legally sufficient or politically fair. It must be revised when new algorithm families, CLI surfaces, or audit profiles are added.

The current draft is framework-level evidence. Before publication-ready status, each row should cite a concrete command transcript, fixture, benchmark, or paper-specific evidence placeholder. Rows for exact optimization and LNS should remain staged until U.16 through U.18 document their implementation surfaces.

5 Conclusion

U.12 turns the algorithm-family roadmap into an operational choice table: select the method family that matches the claim, then carry its output through the declared audit path.

This keeps the family usable as it grows. New methods can be added by stating their claim type, evidence level, CLI surface, and audit handoff rather than by being forced into a single leaderboard.

References

BISECT Project. U.12 algorithm-selection matrix plan, 2026. BISECT internal paper plan.